

Experiment-1

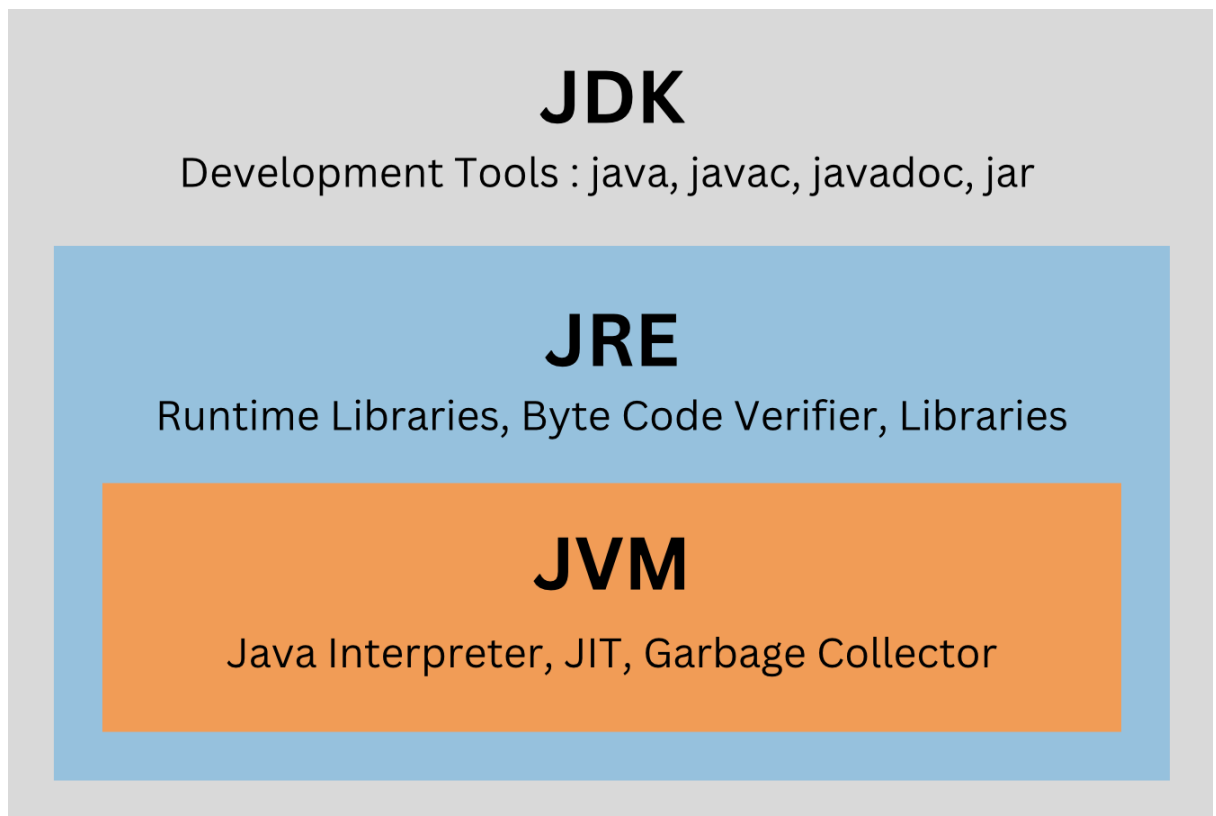
TITLE: Introduction to Java Environment

1. Explore and understand the role of JDK, JRE and JVM.

The Java Development Kit (JDK) is a software development environment that offers a collection of tools and libraries necessary for developing Java-based software applications and applets.

Java Runtime Environment (JRE) provides an environment to run only the Java program onto the system. It is an open-access software distribution that has a Java class library, specific tools, and a separate JVM.

Java Virtual Machine (JVM) is responsible for executing the Java program. JVM is the one that calls the main method present in a Java code. JVM is a part of JRE (Java Runtime Environment).



2. Install latest available JDK and verify the Java Environment.

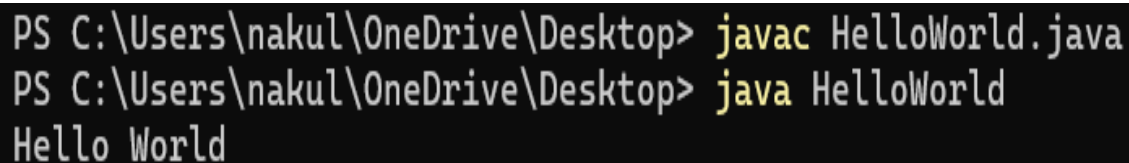
```
C:\Users\nakul>java -version
java version "22.0.1" 2024-04-16
Java(TM) SE Runtime Environment (build 22.0.1+8-16)
Java HotSpot(TM) 64-Bit Server VM (build 22.0.1+8-16, mixed mode, sharing)
```

3. Create a Sample Hello World Program using simple text editor (e.g. Notepad) and show the steps to compile and execute the program using command prompt.

CODE:

```
class HelloWorld
{
    public static void main(String args[])
    {
        System.out.println("Hello World");
    }
}
```

OUTPUT:



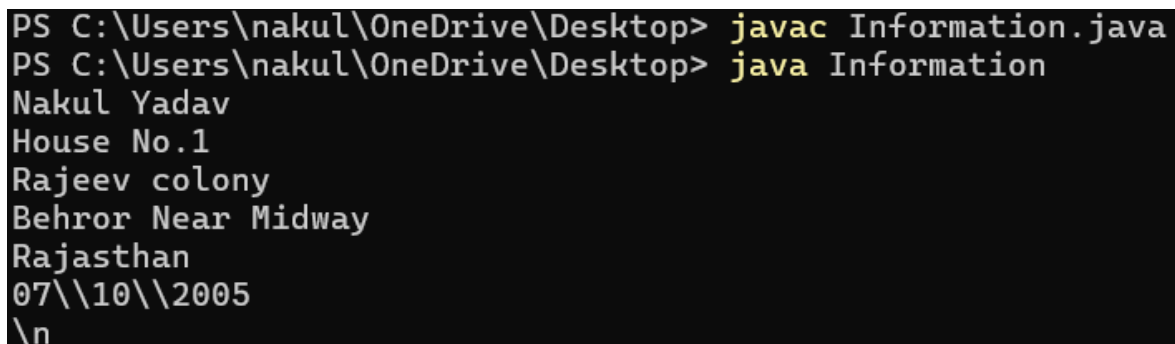
```
PS C:\Users\nakul\OneDrive\Desktop> javac HelloWorld.java
PS C:\Users\nakul\OneDrive\Desktop> java HelloWorld
Hello World
```

4. Display your name and complete address in different lines.

CODE:

```
class Information
{
    public static void main(String args[])
    {
        System.out.println("BHAVYA MITTAL");
        System.out.println("Flat number 306");
        System.out.println("Tower H");
        System.out.println("Kw Shrishti Rajnagar Extension Ghaziabad");
        System.out.println("Uttar Pradesh");
        System.out.println("25\\09\\2005");
        System.out.println("\\n");
    }
}
```

OUTPUT:



```
PS C:\Users\nakul\OneDrive\Desktop> javac Information.java
PS C:\Users\nakul\OneDrive\Desktop> java Information
Nakul Yadav
House No.1
Rajeev colony
Behror Near Midway
Rajasthan
07\\10\\2005
\\n
```

5. Design a visually appealing gradesheet that displays the Name, Roll Number, SAP ID, and Result. Use escape sequences and special characters like '*' to enhance its presentation. [No need to take any input from User].

CODE:

```
class GradeCard
{
public static void main(String args[])
{
    System.out.println("*****");
    System.out.println("\tMy Grade Card");
    System.out.println("Name- Bhavya Mittal");
    System.out.println("Sap Id- 500119005");
    System.out.println("Enrollment number-R2142230023");
    System.out.println("CGPA-: 7.80");
    System.out.println("\tRESULT-PASS");
    System.out.println("*****");
}
}
```

OUTPUT:

```
PS C:\Users\nakul\OneDrive\Desktop> javac GradeCard.java
PS C:\Users\nakul\OneDrive\Desktop> java GradeCard
*****
        My Grade Card
Name- Nakul Yadav
Sap Id- 500121882
Enrollment number-R2142230114
CGPA-: 7.6
        RESULT-PASS
*****
```